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METHOD FOR IMPROVING TRANSIENT RESPONSE FOR POWER CONVERTERS IN PULSE FREQUENCY MODE (PFM)

Abstract

A first input of a modulator circuit is coupled to a current feedback terminal. An output of the modulator circuit is coupled to an input of a power stage circuit. A first input, a second input, and an output of a voltage compensation circuit are coupled to a voltage feedback terminal, a voltage reference terminal, and a compensation terminal, respectively. A first input of a summing circuit is coupled to the output of the voltage compensation circuit. An output of the summing circuit is coupled to a second input of the modulator circuit. An output of an offset generation circuit is coupled to a second input of the summing circuit. An input of a filter circuit is coupled to the output of the voltage compensation circuit. An output of the filter circuit is coupled to an input of the offset generation circuit.

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Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This Application claims the benefit of Provisional Application No. 63/559,503, filed on Feb. 29, 2024, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] Direct current (DC)-to-DC converters and alternating current (AC)-to-DC converters, which may be referred to collectively as power converters, are widely employed in devices of today to perform power conversion. Generally, power converters receive a nominal voltage from a power source, such as a battery, and provide a regulated output voltage at one or more voltage levels. A variety of power converters and topologies can be employed to perform this power conversion. For example, buck converters, boost converters, and buck-boost converters are three basic types of power converter technologies.

SUMMARY

[0003] A circuit includes a power stage circuit, a modulator circuit, a voltage compensation circuit, a summing circuit, an offset generation circuit, and a filter circuit. The power stage circuit has an input and an output. The modulator circuit has a first input, a second input, and an output. The first input of the modulator circuit is coupled to a current feedback terminal. The output of the modulator circuit is coupled to the input of the power stage circuit. The voltage compensation circuit has a first input, a second input, and an output. The first input of the voltage compensation circuit is coupled to a voltage feedback terminal. The second input of the voltage compensation circuit is coupled to a voltage reference terminal. The output of the voltage compensation circuit is coupled to a compensation terminal. The summing circuit has a first input, a second input, and an output. The first input of the summing circuit is coupled to the output of the voltage compensation circuit. The output of the summing circuit is coupled to the second input of the modulator circuit. The offset generation circuit has an input and an output. The output of the offset generation circuit is coupled to the second input of the summing circuit. The filter circuit has an input and an output. The input of the filter circuit is coupled to the output of the voltage compensation circuit. The output of the filter circuit is coupled to the input of the offset generation circuit.

[0004] A circuit includes a voltage-to-current converter, a summing circuit, a current mirror circuit, a current valve circuit, a current source circuit, and a filter circuit. The voltage-to-current converter circuit has an input, a first output, and a second output. The summing circuit has a first input, a second input, and an output. The first input of the summing circuit is coupled to the first output of the voltage-to-current converter circuit. The current mirror circuit has an input and an output. The output of the current mirror circuit is coupled to the second input of the summing circuit. The current valve circuit has an input, an output, and a control terminal. The output of the current valve circuit is coupled to the input of the current mirror circuit. The current source circuit has an output coupled to the input of the current valve circuit. The filter circuit has an input and an output. The input of the filter circuit is coupled to the second output of the voltage-to-current converter circuit. The output of the filter circuit is coupled to the control terminal of the current valve circuit.

[0005] A circuit includes a power stage circuit, a modulator circuit, a voltage compensation circuit, a summing circuit, an offset generation circuit, and a filter circuit. The power stage circuit is configured to output a switching signal based on a modulated signal. The modulator circuit is configured to output the modulated signal based on a comparison between a first feedback signal indicating a feedback current and a target signal indicating a target current peak. The voltage compensation circuit is configured to output a voltage compensation signal based on a comparison between a second feedback signal indicating a feedback voltage and a reference signal indicating a reference voltage. The summing circuit is configured to output the target signal based on the voltage compensation signal and an offset signal. The offset generation circuit is configured to output the offset signal based on a filtered voltage compensation signal. The filter is configured to output the filtered voltage compensation signal based on the voltage compensation signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a circuit diagram of some examples of a system including a voltage converter circuit.

[0007] FIG. **2** is a diagram of some examples of signals corresponding to the system of FIG. **1**.

[0008] FIGS. **3-5**, **6**A, **6**B are circuit diagrams of some other examples of voltage converter circuits similar to the voltage converter circuit of FIG. **1**.

[0009] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (functionally and/or structurally) features.

DETAILED DESCRIPTION

[0010] The following description provides many different examples for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present description. The drawings are not drawn to scale.

[0011] FIG. **1** is a circuit diagram of some examples of a system including a voltage converter circuit **100**. The voltage converter circuit **100** includes a power stage circuit **108**, a modulator circuit **110** and a voltage compensation circuit **114**. The system includes the voltage converter circuit **100**, a voltage source **130**, an inductor **132**, a load circuit **134**, a capacitor **136**, and a current sensing circuit **138**. In the example illustrated in FIG. **1**, the system forms a DC-DC buck converter which generates a voltage across the load circuit **134** that is less than a voltage across the voltage source **130**.

[0012] A first terminal **108***a* of the power stage circuit **108** is coupled to a first terminal **102** of the voltage converter circuit **100**. A second terminal **108***b* of the power stage circuit **108** is coupled to a second terminal **104** of the voltage converter circuit **100**. A third terminal **108***c* of the power stage circuit **108** is coupled to a third terminal **106** of the voltage converter circuit **100**.

[0013] An output **110***c* of the modulator circuit **110** is coupled to a control terminal **108***d* of the power stage circuit **108**. A first input **110***a* of the modulator circuit **110** is coupled to a current feedback terminal **112** of the voltage converter circuit **100**.

[0014] A first input **114***a* of the voltage compensation circuit **114** is coupled to a voltage feedback terminal **116** of the voltage converter circuit **100**. A second input **114***b* of the voltage compensation circuit **114** is coupled to a voltage reference terminal **118**. An output **114***c* of the voltage compensation circuit **114** is coupled to a compensation terminal **120** of the voltage converter circuit **100**.

[0015] A first terminal **130***a* of the voltage source **130** is coupled to terminal **102** of the voltage converter circuit **100**. A second terminal **130***b* of the voltage source **130** is coupled to ground **140**. A first terminal **132***a* of the inductor **132** is coupled to terminal **104** of the voltage converter circuit **100**. A second terminal **132***b* of the inductor **132** is coupled to a first terminal **134***a* of the load circuit **134**. A second terminal **134***b* of the load circuit **134** is coupled to ground **140**. The voltage feedback terminal **116** of the voltage converter circuit **100** is coupled to the first terminal **134***a* of the load circuit **134**. A first terminal **136***a* of the capacitor **136** is coupled to the second terminal **132***b* of the inductor **132** and the first terminal **134***a* of the load circuit **134**. A second terminal **136***b* of the capacitor **136** is coupled to ground **140**. The current sensing circuit **138** is coupled between the second terminal **132***b* of the inductor **132** and the first terminal **134***a* of the load circuit **134**. An output **138***a* of the current sensing circuit **138** is coupled to the current feedback terminal **112** of the voltage converter circuit **100**.

[0016] The voltage compensation circuit **114** receives a second feedback signal FB**2** at input **114***a* from the voltage feedback terminal **116**. The second feedback signal FB**2** indicates a feedback voltage (the voltage across the load circuit **134**). The voltage compensation circuit **114** receives a reference signal REF at input **114***b* from the voltage reference terminal **118**. The reference signal REF indicates a reference voltage. The voltage compensation circuit **114** outputs a voltage compensation signal COMP at output **114***c* based on a comparison between the second feedback signal FB**2** and the reference signal REF.

[0017] The modulator circuit **110** receives a first feedback signal FB**1** at input **110***a* from the current feedback terminal **112**. The first feedback signal FB**1** indicates a feedback current (e.g., the current in the inductor **132** as measured by the current sensing circuit **138**). The modulator circuit **110** receives a target signal TAR at input **110***b*. The target signal TAR indicates a target current peak. The modulator circuit **110** outputs a modulated signal MOD at output **110***c* based on a comparison between the first feedback signal FB**1** and the target signal TAR.

[0018] The power stage circuit **108** receives the modulated signal MOD at control terminal **108***d*. The power stage circuit **108** selectively couples terminal **102** to terminal **104** and selectively couples terminal **104** to terminal **106** based on the modulated signal MOD.

[0019] In some examples, the voltage converter circuit **100** operates in a peak current control mode where the modulator circuit **110** sets the modulated signal MOD to a first value (e.g., a logic "low" value) in response to the first feedback signal FB**1** reaching the target signal TAR. Further, the modulator circuit **110** periodically (e.g., according to a clock signal) sets the modulated signal MOD to a second value (e.g., a logic "high" value).

[0020] In response to the modulated signal MOD having the second value (logic "high"), the power stage circuit **108** couples terminal **102** to terminal **104** and decouples terminal **104** from terminal **106**, thus coupling the voltage source **130** to the load circuit **134** through the inductor **132**. In response, current flows through inductor **132** from terminal **132***a* to terminal **132***b* and to the load circuit **134**.

[0021] In response to the modulated signal MOD having the first value (logic "low"), the power stage circuit **108** decouples terminal **102** from terminal **104** and couples terminal **104** to terminal **106**, thus decoupling the voltage source **130** from the inductor **132** and coupling the inductor **132** to ground **140**. In response, inductor **132** generates a current which flows from terminal **132***a* to terminal **132***b* and to the load circuit **134**.

[0022] The power stage circuit **108** performs this switching (coupling and decoupling between terminals **102**, **104**, **106**) at relatively high frequencies. Consequently, switching losses in the power stage circuit **108** are increased. These increased switching losses cause the voltage converter circuit **100** to suffer from reduced efficiency when operating at lighter loads (e.g., when the current demand from the load circuit **134** is decreased).

[0023] To improve the efficiency of the voltage converter circuit **100** at lighter loads, the voltage converter circuit **100** includes a target control circuit **122** coupled between the output **114***c* of the voltage compensation circuit **114** and input **110***b* of the modulator circuit **110**. An input **122***a* of the target control circuit **122** is coupled to the output **114***c* of the voltage compensation circuit **114**. An output **122***b* of the target control circuit **122** is coupled to input **110***b* of the modulator circuit **110**.

[0024] The target control circuit **122** receives the voltage compensation signal COMP at input **122***a* and outputs the target signal TAR at output **122***b* based on the voltage compensation signal COMP. In response to the load falling below a load threshold, the target control circuit **122** causes the voltage converter circuit **100** to switch from peak current control mode to frequency control mode. For example, in response to the load falling below a load threshold, the target control circuit **122** sets the target signal TAR to a controllable reference target value. This causes the power stage circuit **108** to perform the switching (coupling and decoupling between terminals **102**, **104**, **106**) at a controllable reference target frequency, independent of voltage compensation signal COMP. Conversely, in response to the load rising above the load threshold, the target control circuit **122** causes the voltage converter circuit **100** to switch from frequency control mode to peak current control mode. For example, in response to the load rising above the load threshold, the target control circuit **122** sets the target signal TAR based on the voltage compensation signal COMP. By reducing the reference target frequency when operating in frequency control mode (during lighter loads), the switching losses in the power stage circuit **108** can be reduced. Thus, efficiency can be improved during lighter loads.

[0025] The target control circuit **122** includes a summing circuit **124** and an offset generation circuit **126**. A first input **124***a* of the summing circuit **124** is coupled to output **114***c* of the voltage compensation circuit **114**. A second input **124***b* of the summing circuit **124** is coupled to an output **126***b* of the offset generation circuit **126**. An output **124***c* of the summing circuit **124** is coupled to input **110***b* of the modulator circuit **110**.

[0026] The summing circuit **124** receives the voltage compensation signal COMP at input **124***a* and an offset signal OFF at input **124***b*. The summing circuit **124** outputs the target signal TAR at output **124***c* based on a sum of the compensation signal COMP and the offset signal OFF.

[0027] In some examples, the offset generation circuit **126** receives the voltage compensation signal COMP at input **126***a* and outputs the offset signal OFF based on the voltage compensation signal COMP. However, in such examples, the voltage converter circuit **100** may exhibit reduced performance during transitions between peak current control mode and frequency control mode.

[0028] For example, as shown in FIG. **2**, in response to the load decreasing, the first feedback signal FB**1** (indicating feedback current) decreases and the second feedback signal FB**2** (indicating feedback voltage) increases. In response to the second feedback signal FB**2** increasing, the voltage compensation signal COMP decreases. In examples where the offset generation circuit **126** outputs the offset signal OFF based on the voltage compensation signal COMP, the offset signal OFF increases in response to the voltage compensation signal COMP reaching (e.g., falling to or below) a compensation threshold TH.sub.COMP, as shown by dashed line **202**, so the target signal TAR (the sum of the offset signal OFF and the compensation signal COMP) does not fall below the reference target value T.sub.ref, as shown by dashed line **204**. Because the target signal TAR is prevented from falling below the reference target value T.sub.ref (despite the voltage compensation signal COMP continuing to fall after the target signal TAR reaches the reference target value T.sub.ref), the voltage converter circuit **100** takes longer to correct the second feedback signal FB**2** at this transition from peak current control mode to frequency control mode. As a result, the second feedback signal FB**2** has increased overshoot at the transition, as shown by dashed line **206**.

[0029] Further, as shown in FIG. **2**, in response to the load increasing, the first feedback signal FB**1** increases and the second feedback signal FB**2** decreases. In response to the second feedback signal FB**2** decreasing, the voltage compensation signal COMP increases. In examples where the offset generation circuit **126** outputs the offset signal OFF based on the voltage compensation signal COMP, the offset signal OFF decreases in response to the voltage compensation signal COMP increasing, as shown by dashed line **208**. Because the offset signal OFF decreases as the voltage compensation signal COMP increases, the target signal TAR (the sum of the offset signal OFF and the voltage compensation signal COMP) remains approximately unchanged until the voltage compensation signal COMP reaches the compensation threshold TH.sub.COMP (and offset signal OFF reaches zero), as shown by dashed line **210**. Only after the voltage compensation signal COMP reaches the compensation threshold TH.sub.COMP (and offset signal OFF reaches zero) does the target signal TAR begin substantially increasing, as shown by dashed line **212**. Because of this delay in the target signal TAR response to the change in the voltage compensation signal COMP, the voltage converter circuit **100** takes longer to correct the second feedback signal FB**2** at this transition from frequency control mode to peak current control mode. As a result, the second feedback signal FB**2** has increased undershoot at the transition, as shown by dashed line **214**.

[0030] To improve performance during the transitions between peak current control mode and frequency control mode, the target control circuit **122** of the voltage converter circuit **100** of FIG. **1** includes a filter circuit **128** coupled between the voltage compensation circuit **114** and the offset generation circuit **126**. The filter circuit **128** is or includes a low pass filter. An input **128***a* of the filter circuit **128** is coupled to the output **114***c* of the voltage compensation circuit **114**. An output **128***b* of the filter circuit **128** is coupled to the input **126***a* of the offset generation circuit **126**.

[0031] The filter circuit **128** receives the voltage compensation signal COMP at input **128***a* and outputs the filtered voltage compensation signal FIL at output **128***b* based on the voltage compensation signal COMP. The offset generation circuit **126** receives the filtered voltage compensation signal FIL at input **126***a* and outputs the offset signal OFF at output **126***b* based on the filtered voltage compensation signal FIL. The filter circuit **128** adds a delay at the input **126***a* of the offset generation circuit **126** to improve the performance of the voltage converter circuit **100** during the transitions between the control modes.

[0032] For example, as shown in FIG. **2**, in response to the load decreasing, the first feedback signal FB**1** (indicating feedback current) decreases and the second feedback signal FB**2** (indicating feedback voltage) increases. In response to the second feedback signal FB**2** increasing, the voltage compensation signal COMP decreases. In response to the voltage compensation signal COMP decreasing, the filtered voltage compensation signal FIL decreases (at a reduced rate compared to voltage compensation signal COMP). In response to the filtered voltage compensation signal FIL reaching (e.g., falling to or below) the compensation threshold TH.sub.COMP, the offset signal OFF increases. Because the filtered voltage compensation signal FIL decreases at a reduced rate (compared to the voltage compensation signal COMP), the filtered voltage compensation signal FIL reaches the compensation threshold TH.sub.COMP after the voltage compensation signal COMP is substantially below the compensation threshold TH.sub.COMP (e.g., when the voltage compensation signal COMP is approximately zero). Because of this delay in the filtered voltage compensation signal FIL reaching the compensation threshold TH.sub.COMP, the target signal TAR (the sum of the offset signal OFF and the voltage compensation signal COMP) falls substantially below the reference target value T.sub.ref. Because the target signal TAR decreases according to the decrease in the voltage compensation signal COMP during this transition from peak current control mode to frequency control mode, the voltage converter circuit **100** corrects the second feedback signal FB**2** more quickly at the transition. As a result, the second feedback signal FB**2** has reduced overshoot at the transition. For example, the overshoot may be reduced by about 5 percent or more. By reducing the overshoot in the second feedback signal FB**2** (which indicates the voltage across the load circuit **134**), voltage stress on the system can be reduced, reliability of the system can be improved, and functional safety of the system can be improved.

[0033] Further, as shown in FIG. **2**, in response to the load increasing, the first feedback signal FB**1** increases and the second feedback signal FB**2** decreases. In response to the second feedback signal FB**2** decreasing, the voltage compensation signal COMP increases. In response to the voltage compensation signal COMP increasing, the filtered voltage compensation signal FIL increases (at a reduced rate relative to the voltage compensation signal COMP). In response to the filtered voltage compensation signal FIL increasing at a reduced rate, the offset signal OFF decreases at a reduced rate. Because the decrease in the offset signal OFF is slower than the increase in the voltage compensation signal COMP, the target signal TAR (the sum of the offset signal OFF and the voltage compensation signal COMP) increases as the voltage compensation signal COMP increases. Thus, the delay illustrated by dashed line **210** can be reduced or eliminated. By reducing this delay, the voltage converter circuit **100** can correct the second feedback signal FB**2** more quickly at this transition from frequency control mode to peak current control mode. Thus, the second feedback signal FB**2** has reduced undershoot at the transition. For example, the undershoot may be reduced by about 6 percent or more. By reducing undershoot in the second feedback signal FB**2** (which indicates the voltage across the load circuit **134**), the reliability of the system can be improved. Further, the size of output capacitor **136** can be reduced and thus the cost of the system may be reduced.

[0034] Although FIG. **1** illustrates a buck converter, it will be appreciated that in some examples, the system may alternatively be configured as a boost converter, a buck-boost converter, or the like. In some examples, the system may be configured as a multi-phase DC-DC converter including a plurality of voltage converter circuits coupled to a corresponding plurality of inductors.

[0035] In some examples, the system is an automotive system where the voltage source **130** is a car battery (e.g., a 12 volt battery) and the load circuit **134** is a vehicle infotainment device, a vehicle instrument cluster, a vehicle stereo, an Advanced driver-assistance systems (ADAS), a sensor fusion system, a vehicle radar system, a motor drive system, or the like. In some examples, the voltage converter circuit **100** is formed on a single monolithic integrated chip.

[0036] FIG. **3** is a circuit diagram of some examples of a voltage converter circuit **300** similar to the voltage converter circuit **100** of FIG. **1**.

[0037] The target control circuit **122** includes a voltage-to-current converter circuit **310**. An input **310***a* of the voltage-to-current converter circuit **310** is coupled to the output **114***c* of the voltage compensation circuit **114**. A first output **310***b* of the voltage-to-current converter circuit **310** is coupled to input **124***a* of the summing circuit **124**. A second output **310***c* of the voltage-to-current converter circuit **310** is coupled to the input **128***a* of the filter circuit **128**.

[0038] The offset generation circuit **126** includes a current source circuit **302**, a current valve circuit **304**, and a current mirror circuit **306**. The current source circuit **302** is coupled to a voltage source **312**. An output **302***a* of the current source circuit **302** is coupled to an input **304***a* of the current valve circuit **304**. A control terminal **304***c* of the current valve circuit **304** is coupled to the output **128***b* of the filter circuit **128**. An output **304***b* of the current valve circuit **304** is coupled to an input **306***a* of the current mirror circuit **306**. An output **306***b* of the current mirror circuit **306** is coupled to input **124***b* of the summing circuit **124**.

[0039] The target control circuit **122** includes a slope compensation circuit **308**. An output **308***a* of the slope compensation circuit **308** is coupled to a third input **124***d* of the summing circuit **124**.

[0040] The voltage-to-current converter circuit **310** receives the voltage compensation signal at input **310***a* from the output **114***c* of voltage compensation circuit **114**. The voltage-to-current converter circuit **310** outputs a converted voltage compensation signal at output **310***b* (e.g., converted from a voltage signal to a current signal) based on the voltage compensation signal. In addition, the voltage-to-current converter circuit **310** outputs an amplified voltage compensation signal at output **310***c* based on the voltage compensation signal.

[0041] The filter circuit **128** receives the amplified voltage compensation signal at input **128***a* and outputs the filtered voltage compensation signal at output **128***b* based on the amplified voltage compensation signal and a cutoff frequency of the filter circuit **128**.

[0042] The current source circuit **302** outputs a reference target current signal at output **302***a*. The current valve circuit **304** receives the reference target current signal at input **304***a* and receives the filtered voltage compensation signal at control terminal **304***c*. The current valve circuit **304** outputs an offset current signal at output **304***b*. The offset current signal has a magnitude equal to a portion of the magnitude of the reference target current signal. The portion is based on the filtered voltage compensation signal. For example, the portion increases in response to the filtered voltage compensation signal increasing, and the portion decreases in response to the filtered voltage compensation signal decreasing. The current mirror circuit **306** receives the offset current signal at input **306***a* and outputs a copy of the offset current signal at output **306***b*.

[0043] The slope compensation circuit **308** outputs a slope compensation ramp signal at output **308***a*. The summing circuit **124** receives the copy of the offset current signal at input **124***b*, the converted voltage compensation signal at input **124***a*, and the slope compensation ramp signal at input **124***d*. The summing circuit **124** outputs the target signal based on a sum of the copy of the offset current signal, the converted voltage compensation signal, and the slope compensation ramp signal.

[0044] FIG. **4** is a circuit diagram of some examples of a voltage converter circuit **400** similar to the voltage converter circuit **300** of FIG. **3**.

[0045] The target control circuit **122** includes a current-to-voltage converter circuit **402**. An input **402***a* of the current-to-voltage converter circuit **402** is coupled to the output **124***c* of the summing circuit **124**. An output **402***b* of the current-to-voltage converter circuit **402** is coupled to input **110***b* of the modulator circuit **110**.

[0046] The circuit **400** includes a current sense amplifier circuit **404**. A first input **404***a* of the current sense amplifier circuit **404** is coupled to current feedback terminal **112**. A second input **404***b* of the current sense amplifier circuit **404** is coupled to a second current feedback terminal **406**. An output **404***c* of the current sense amplifier circuit **404** is coupled to input **110***a* of the modulator circuit **110**. In some examples, the current sense amplifier circuit **404** is coupled to a current sense resistor (not shown) which is coupled between an inductor (e.g., **132** of FIG. **1**) and a load circuit (e.g., **134** of FIG. **1**). For example, input **404***a* is coupled to a first terminal of the current sense resistor (through terminal **112**) and input **404***b* is coupled to a second terminal of the current sense resistor (through terminal **406**).

[0047] The current-to-voltage converter circuit **402** converts the target signal from a current signal to a voltage signal. The current sense amplifier circuit **404** converts the first feedback signal from a current signal to a voltage signal.

[0048] The modulator circuit **110** includes a comparator circuit **408** and a logic circuit **410**. A first input **408***a* of the comparator circuit **408** is coupled to the output **404***c* of the current sense amplifier circuit **404**. A second input **408***b* of the comparator circuit **408** is coupled to the output **402***b* of the current-to-voltage converter circuit **402**. A first input **410***a* of the logic circuit **410** is coupled to an output **408***c* of the comparator circuit **408**. A second input **410***b* of the logic circuit **410** is coupled to a clock terminal **418** of the voltage converter circuit **400**.

[0049] The comparator circuit **408** receives the target signal at input **408***b* and the first feedback signal at input **408***a*. The comparator circuit **408** outputs a comparator output signal at output **408***c* based on a comparison between the target signal and the first feedback signal. The logic circuit **410** receives the comparator output signal at input **410***a* and a clock signal at input **410***b*. The logic circuit **410** outputs the modulated signal at output **410***c* based on the comparator output signal, the clock signal, and the logic of the logic circuit **410** (e.g., a logic table or the like).

[0050] The power stage circuit **108** includes a transistor driver circuit **412**. An input **412***a* of the transistor driver circuit **412** is coupled to an output **410***c* of the logic circuit **410**. In some examples, the power stage circuit **108** includes transistors **414**, **416**. A first terminal **414***a* of transistor **414** is coupled to terminal **102**. A second terminal **414***b* of transistor **414** and a first terminal **416***a* of transistor **416** are coupled to terminal **104**. A second terminal **416***b* of transistor **416** is coupled to terminal **106**. A control terminal **414***c* of transistor **414** is coupled to a first output **412***b* of the transistor driver circuit **412**. A control terminal **416***c* of transistor **416** is coupled to a second output **412***c* of the transistor driver circuit **412**.

[0051] The transistor driver circuit **412** receives the modulated signal at input **412***a*. The transistor driver circuit **412** outputs a first driver signal at output **412***b* and a second driver signal at output **412***c*. The driver signals control transistors **414**, **416** which selectively couple terminals **102**, **104**, **106** based on the driver signals.

[0052] In some examples, transistors **414**, **416** are external to voltage converter circuit **400**. In such examples, output **412***b* is coupled to terminal **102**, output **412***c* is coupled to terminal **104**, and terminal **106** is omitted from the voltage converter circuit **400**. In such examples, transistors **414**, **416** may be off-chip (on a separate chip) and control terminals **414***c*, **416***c* may be coupled to terminals **102**, **104**, respectively, by wiring that is external to the chips.

[0053] FIG. **5** is a circuit diagram of some examples of a voltage converter circuit **500** similar to the voltage converter circuit **400** of FIG. **4**.
[0054] The voltage compensation circuit **114** includes an error amplifier **502**. A first input **502***a* of the error amplifier **502** is coupled to the feedback voltage terminal **116**. A second input **502***b* of the error amplifier **502** is coupled to an output **508***a* of a reference voltage generation circuit **508**. The output **502***c* of the error amplifier **502** is coupled to the input **310***a* of the voltage-to-current converter circuit **310** and the compensation terminal **120**.
[0055] The voltage compensation circuit **114** includes an input impedance circuit **504** coupled between the feedback voltage terminal **116** and input **502***a* of the error amplifier **502**. For example, a first terminal **504***a* if the input impedance circuit **504** is coupled to the voltage feedback terminal **116** and a second terminal **504***b* of the input impedance circuit **504** is coupled to the first input **502***a* of the error amplifier.
[0056] The voltage compensation circuit **114** includes a feedback impedance circuit **506** coupled between input **502***a* of the error amplifier **502** and the output **502***c* of the error amplifier **502**. For example, a first terminal **506***a* of the feedback impedance circuit **506** is coupled to the first input **502***a* of the error amplifier **502** and a second terminal **506***b* of the feedback impedance circuit **506** is coupled to the output **502***c* of the error amplifier **502**.
[0057] A transfer function of the voltage compensation circuit **114** has at least one pole and at least one zero. The low pass filter of the filter circuit **128** has a time constant corresponding to a time delay. The time constant of the filter circuit **128** and the zero of the transfer function of the voltage compensation circuit **114** are approximately equal to optimize the delay generated by the filter circuit **128**. In some cases, if the time constant is less than the zero and thus the delay is too short, the rate of change of the filtered voltage compensation signal may not be slow enough and thus the second feedback signal may have poor overshoot/undershoot at the transitions. If the time constant is too high and thus the delay is too long, the rate of change of the filtered voltage compensation signal may be too slow, the delay may be longer than necessary and thus performance may be reduced. By setting the time constant of the filter circuit **128** and the zero of the transfer function of the voltage compensation circuit **114** to be approximately equal, the length of the delay at the filter circuit **128** may be optimized according to the performance of the voltage compensation circuit **114** (e.g., the delay can be long enough but not too long).
[0058] FIG. **6**A and FIG. **6**B are a circuit diagram of some examples of a voltage converter circuit **600** similar to the voltage converter circuit **500** of FIG. **5**.
[0059] The voltage-to-current converter circuit **310** includes an amplifier **602**, a transistor **604**, and a resistor **606**. A first terminal **602***a* of the amplifier **602** is coupled to the output **502***c* of the error amplifier **502**. A second terminal **602***b* of the amplifier **602** is coupled to a second terminal **604***b* of transistor **604** and a first terminal **606***a* of resistor **606**. An output **602***c* of the amplifier **602** is coupled to a control terminal **604***c* of transistor **604** and the input **128***a* of the filter circuit **128**. A first terminal **604***a* of transistor **604** is coupled to input **124***a* of the summing circuit **124**. A second terminal **606***b* of resistor **606** is coupled to ground **630**. Transistor **604** passes a current from terminal **406***a* to terminal **604***b*. The magnitude of the current is based on the voltage at terminal **602***a*.
[0060] The filter circuit **128** includes a resistor **608** and a capacitor **610**. A first terminal **608***a* of resistor **608** is coupled to the output **602***c* of amplifier **602**. A second terminal **608***b* of resistor **608** is coupled to a first terminal **610***a* of capacitor **610** and the control terminal **304***c* of the current valve circuit **304**. A second terminal **610***b* of capacitor **610** is coupled to ground **630**.
[0061] The current valve circuit **304** includes a transistor **612** and a resistor **614**. A first terminal **612***a* of transistor **612** is coupled to the output **302***a* of the current source circuit **302** and the input **306***a* of the current mirror circuit **306**. A second terminal **612***b* of transistor **612** is coupled to a first terminal **614***a* of resistor **614**. A control terminal **612***c* of transistor **612** is coupled to terminal **608***b* of resistor **608** and terminal **610***a* of capacitor **610**. A second terminal **614***b* of resistor **614** is

coupled to ground **630**. A first portion of the current from output by the current source circuit **302** is output to the current mirror circuit **306**. Transistor **612** passes a second portion of the current from the current source circuit **302** to ground **630** based on the magnitude of the voltage at terminal **608***b* of resistor **608** and terminal **610***a* of capacitor **610**.

[0062] The current mirror circuit **306** includes a first transistor **616** and a second transistor **618**. A first terminal **616***a* of transistor **616** is coupled to the output **302***a* of the current source circuit **302** and the first terminal **612***a* of transistor **612**. A second terminal **616***b* of transistor **616** is coupled to ground **630**. A control terminal **616***c* of transistor **616** is coupled to a control terminal **618***c* of transistor **618** and to terminal **616***a* of transistor **616**. A first terminal **618***a* of transistor **618***a* is coupled to input **124***b* of summing circuit **124**. A second terminal **618***b* of transistor **618** is coupled to ground **630**.

[0063] The summing circuit shorts input **124***a*, input **124***b*, and input **124***d* to output **124***c*. For example, terminal **618***a* of transistor **618**, terminal **604***a* of transistor **604**, and output **308***a* of the slope compensation circuit **308** are coupled together and to input **408***b* of the comparator circuit **408**.

[0064] The current-to-voltage converter circuit **402** includes a resistor **620** and a voltage terminal **632** (e.g., a voltage supply terminal or a ground terminal). A first terminal **620***a* of resistor **620** is coupled to terminal **618***a* of transistor **618**, terminal **604***a* of transistor **604**, output **308***a* of the slope compensation circuit **308**, and input **408***b* of comparator circuit **408**. A second terminal **620***b* of resistor **620** is coupled to the voltage terminal **632**.

[0065] The input impedance circuit **504** includes a resistor **622** having a first terminal **622***a* coupled to terminal **116** and a second terminal **622***b* coupled to input **502***a* of the error amplifier **502**.

[0066] The feedback impedance circuit **506** includes a resistor **624**, a capacitor **626**, and a capacitor **628**. A first terminal **624***a* of resistor **624** and a first terminal **628***a* of capacitor **628** are coupled to input **502***a* of error amplifier **502**. A second terminal **624***b* of resistor **624** is coupled to a first terminal **626***a* of capacitor **626**. A second terminal **626***b* of capacitor **626** and a second terminal **628***b* of capacitor **628** are coupled to the output **502***c* of error amplifier **502**.

[0067] As described with regard to FIG. **5**, the time constant of the filter circuit **128** is approximately equal to the zero of the transfer function of the voltage compensation circuit **114** to optimize the delay caused by the filter circuit **128**. In other words, the cutoff frequency of the filter circuit **128** is approximately equal to the zero frequency of the transfer function of the voltage compensation circuit **114**. This can be achieved in the circuit **600** of FIG. **6**A and FIG. **6**B by making the product of the resistance of resistor **608** and the capacitance of capacitor **610** approximately equal to the product of the resistance of resistor **624** and the capacitance of capacitor **626**.

[0068] Although the voltage compensation circuit **114** of FIG. **6**A and FIG. **6**B includes internal compensation where the input impedance circuit **504** and the feedback impedance circuit **506** are internal to the voltage converter circuit **600** (e.g., the input impedance circuit **504** and the feedback impedance circuit **506** are on the same chip as the other components of circuit **600**), it will be appreciated that in some other examples, the voltage compensation circuit **114** may include external compensation where the input impedance circuit **504** and the feedback impedance circuit **506** are external to the voltage converter circuit **600** (e.g., the input impedance circuit **504** and the feedback impedance circuit **506** are a different chip than the other components of circuit **600**). In such examples, the feedback impedance circuit **506** is coupled to the compensation terminal **120** and the input impedance circuit **504** is coupled to the voltage feedback terminal **116** by external wiring. In such examples, resistor **608** and capacitor **610** are variable so that their resistance and capacitance, respectively, can be adjusted (e.g., via an additional terminal of the circuit **600**) based on resistance and capacitance of the external feedback impedance circuit in order to optimize the delay of the filter circuit **128** according to the external feedback impedance.

[0069] Although the voltage compensation (of the zeros and poles) circuit **114** of FIG. **6**A and FIG.

6B includes a type II compensator, it will be appreciated that voltage compensation circuit **114** could alternatively include a type III compensator or some other type of compensator.
[0070] The methods are illustrated and described above as a series of acts or events, but the illustrated ordering of such acts or events is not limiting. For example, some acts or events may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Also, some illustrated acts or events are optional to implement one or more aspects or embodiments of this description. Further, one or more of the acts or events depicted herein may be performed in one or more separate acts and/or phases. In some embodiments, the methods described above may be implemented in a computer readable medium using instructions stored in a memory.
[0071] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.
[0072] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or re-configurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.
[0073] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.
[0074] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.
[0075] While the use of particular transistors are described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a metal-oxide-silicon FET (“MOSFET”) (such as an n-channel MOSFET, nMOSFET, or a p-channel MOSFET, pMOSFET), a bipolar junction transistor (BJT—e.g. NPN or PNP), insulated gate bipolar transistors (IGBTs), and/or junction field effect transistor (JFET) may be used in place of or in conjunction with the devices disclosed herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other type of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).
[0076] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated

circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0077] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. Unless otherwise stated, “about,” “approximately,” or “substantially” preceding a value means +/−10 percent of the stated value, or, if the value is zero, a reasonable range of values around zero. Modifications are possible in the described examples, and other implementations are possible, within the scope of the claims.

Claims

1. A circuit comprising: a power stage circuit having an input and an output; a modulator circuit having a first input, a second input, and an output, the first input of the modulator circuit coupled to a current feedback terminal, the output of the modulator circuit coupled to the input of the power stage circuit; a voltage compensation circuit having a first input, a second input, and an output, the first input of the voltage compensation circuit coupled to a voltage feedback terminal, the second input of the voltage compensation circuit coupled to a voltage reference terminal, the output of the voltage compensation circuit coupled to a compensation terminal; a summing circuit having a first input, a second input, and an output, the first input of the summing circuit coupled to the output of the voltage compensation circuit, the output of the summing circuit coupled to the second input of the modulator circuit; an offset generation circuit having an input and an output, the output of the offset generation circuit coupled to the second input of the summing circuit; and a filter circuit having an input and an output, the input of the filter circuit coupled to the output of the voltage compensation circuit, the output of the filter circuit coupled to the input of the offset generation circuit.

2. The circuit of claim 1, the offset generation circuit including: a current source having an output; and a current valve circuit having an input, an output, and a control terminal, the input of the current valve circuit coupled to the output of the current source, the output of the current valve circuit coupled to the second input of the summing circuit, the control terminal of the current valve circuit coupled to the output of the filter circuit.

3. The circuit of claim 2, the offset generation circuit further including: a current mirror circuit having an input and an output, the input of the current mirror circuit coupled to output of the current valve circuit, the output of the current mirror circuit coupled to the second input of the summing circuit.

4. The circuit of claim 1, the summing circuit having a third input, the circuit further comprising: a slope compensation circuit having an output coupled to the third input of the summing circuit.

5. The circuit of claim 1, the voltage compensation circuit including: an error amplifier having a first input, a second input, and an output, the second input of the error amplifier coupled to the voltage reference terminal, the output of the error amplifier coupled to the input of the filter circuit, an input impendence circuit having a first terminal coupled to the voltage feedback terminal and a second terminal coupled to the first input of the error amplifier, and a feedback impedance circuit having a first terminal coupled to the first input of the error amplifier and a second terminal coupled to the output of the error amplifier, the filter circuit including a low pass filter, wherein a transfer function of the voltage compensation circuit has a pole and a zero, wherein the low pass filter has a time constant, and wherein the time constant is approximately equal to the zero.

6. The circuit of claim 1, further comprising: a voltage-to-current converter circuit having an input, a first output, and second output, the input coupled to the output of the voltage compensation circuit, the first output of the voltage-to-current converter circuit coupled to the first input of the summing circuit, the second output of the voltage-to-current converter circuit coupled to the input of the filter circuit.

7. The circuit of claim 6, wherein the current feedback terminal is a first current feedback terminal, the circuit further comprising: a current sense amplifier circuit having a first input, a second input, and an output, the first input of the current sense amplifier circuit coupled to the first current feedback terminal, the second input of the current sense amplifier circuit coupled to a second current feedback terminal, the output of the current sense amplifier circuit coupled to the first input of the modulator circuit; and a current-to-voltage converter circuit having an input and an output, the input of the current-to-voltage converter circuit coupled to the output of the summing circuit, the output of the current-to-voltage converter circuit coupled to the second input of the modulator circuit.

8. The circuit of claim 7, the modulator circuit including: a comparator circuit having a first input, a second input, and an output, the first input of the comparator circuit coupled to the output of the current sense amplifier circuit, the second input of the comparator circuit coupled to the output of the current-to-voltage converter circuit; and a logic circuit having an input and an output, the input of the logic circuit coupled to the output of the comparator circuit, the output of the logic circuit coupled to the input of the power stage circuit.

9. The circuit of claim 8, the power stage circuit including: a transistor driver circuit having an input and an output, the input of the transistor driver circuit coupled to the output of the logic circuit.

10. A circuit comprising: a voltage-to-current converter circuit having an input, a first output, and a second output; a summing circuit having a first input, a second input, and an output, the first input of the summing circuit coupled to the first output of the voltage-to-current converter circuit; a current mirror circuit having an input and an output, the output of the current mirror circuit coupled to the second input of the summing circuit; a current valve circuit having an input, an output, and a control terminal, the output of the current valve circuit coupled to the input of the current mirror circuit; a current source circuit having an output coupled to the input of the current valve circuit; and a filter circuit having an input and an output, the input of the filter circuit coupled to the second output of the voltage-to-current converter circuit, the output of the filter circuit coupled to the control terminal of the current valve circuit.

11. The circuit of claim 10, the summing circuit having a third input, the circuit further comprising: a slope compensation circuit having an output coupled to the third input of the summing circuit.

12. The circuit of claim 11, the filter circuit including: a first resistor having a first terminal and a second terminal, the first terminal of the first resistor coupled to the second output of the voltage-to-current converter circuit; and a capacitor having a first terminal and a second terminal, the first terminal of the capacitor coupled to the second terminal of the first resistor and the control terminal of the current valve circuit, the second terminal of the capacitor coupled to ground.

13. The circuit of claim 12, the voltage-to-current converter circuit including: an amplifier having a first input, a second input, and an output, the first terminal of the amplifier coupled to the input of the voltage-to-current converter circuit; a first transistor having a first terminal, a second terminal, and a control terminal, the first terminal of the first transistor coupled to the first input of the summing circuit, the second terminal of the first transistor coupled to the second input of the amplifier, the control terminal of the first transistor coupled to the output of the amplifier and the first terminal of the first resistor; and a second resistor having a first terminal and a second terminal, the first terminal of the second resistor coupled to the second terminal of the first transistor, the second terminal of the first resistor coupled to ground.

14. The circuit of claim 13, the current valve circuit including: a second transistor having a first terminal, a second terminal, and a control terminal, the first terminal of the second transistor coupled to the output of the current source circuit and the input of the current mirror circuit, the control terminal of the second transistor coupled to the second terminal of the first resistor and the first terminal of the capacitor; and a third resistor having a first terminal and a second terminal, the first terminal of the third resistor coupled to the second terminal of the second transistor, the second terminal of the third resistor coupled to ground.

15. The circuit of claim 14, the current mirror circuit including: a third transistor having a first terminal, a second terminal, and a control terminal, the first terminal of the third transistor coupled to the first terminal of the second transistor and the output of the current source circuit, the second terminal of the third transistor coupled to ground, the control terminal of the third transistor coupled to the first terminal of the third transistor; and a fourth transistor having a first terminal, and second terminal, and a control terminal, the first terminal of the fourth transistor coupled to the second input of the summing circuit, the second terminal of the fourth transistor coupled to ground, the control terminal of the fourth transistor coupled to the control terminal of the third transistor.

16. The circuit of claim 15, the first terminal of the fourth transistor coupled to the first terminal of the first transistor and the output of the slope compensation circuit.

17. The circuit of claim 16, further comprising: a fourth resistor having a first terminal and a second terminal, the first terminal of the fourth resistor coupled to the first terminal of the fourth transistor, the first terminal of the first transistor, and the output of the slope compensation circuit, the second terminal of the fourth resistor coupled to a voltage supply terminal.

18. A circuit comprising: a power stage circuit configured to output a switching signal based on a modulated signal; a modulator circuit configured to output the modulated signal based on a comparison between a first feedback signal indicating a feedback current and a target signal indicating a target current peak; a voltage compensation circuit configured to output a voltage compensation signal based on a comparison between a second feedback signal indicating a feedback voltage and a reference signal indicating a reference voltage; a summing circuit configured to output the target signal based on the voltage compensation signal and an offset signal; an offset generation circuit configured to output the offset signal based on a filtered voltage compensation signal; and a filter circuit configured to output the filtered voltage compensation signal based on the voltage compensation signal.

19. The circuit of claim 18, further comprising: a voltage-to-current converter circuit configured to output a converted voltage compensation signal and an amplified voltage compensation signal based on the voltage compensation signal, wherein the summing circuit is configured to output the target signal based on a sum of the converted voltage compensation signal and the offset signal, wherein the filter circuit is configured to output the filtered voltage compensation signal based on the amplified voltage compensation signal.

20. The circuit of claim 19, further comprising: a slope compensation circuit configured to output a slope compensation ramp signal, wherein the summing circuit is configured to output the target signal based on a sum of the converted voltage compensation signal, the offset signal, and the slope compensation ramp signal.